THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): The law of constant proportions is true for compounds but not for mixtures. Reason (R): In a compound the elements are chemically combined in a fixed ratio. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A compound of A and B contains 75 per cent A by mass. A second compound of the same elements contains 60 per cent A. The ratio of the masses of A that combine with a fixed mass of B in the two compounds is: 1

(a) 4 : 3 **(b)** 2 : 1 **(c)** 3 : 2 **(d)** 3 : 1

3. A student heats 10 g of CaCO3. Only 5.6 g of solid CaO is left. The mass of CO2 evolved is: 1

(a) 5.6 g **(b)** 15.6 g **(c)** 10 g **(d)** 4.4 g

4. Assertion (A): The mass of a gas-releasing reaction in an open vessel appears to decrease. Reason (R): Mass is destroyed during reactions in open vessels. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A sample of a compound contains 2.4 g of carbon and 6.4 g of oxygen. The ratio C : O by mass and the likely formula (C = 12, O = 16) are: 1

(a) 3 : 16, CO_3 **(b)** 3 : 8, CO_2 **(c)** 1 : 2, CO_2 **(d)** 3 : 4, CO

6. An element X forms a chloride XCl_2 with 52.5 per cent X by mass? (Cl = 35.5) The atomic mass of X is closest to: 1

(a) 56 **(b)** 24 **(c)** 40 **(d)** 78.5

7. Assertion (A): Sodium chloride conducts electricity in the molten state but not in the solid state. Reason (R): In the molten state, the ions are free to move. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. An oxide of nitrogen contains 63.6 per cent nitrogen. The empirical formula (N = 14, O = 16) is: 1

(a) N_2O **(b)** N_2O_3 **(c)** NO **(d)** NO_2

9. The simplest ratio of atoms in a compound with 40 per cent C, 6.7 per cent H, 53.3 per cent O is: 1

(a) 2 : 1 : 1 **(b)** 1 : 1 : 1 **(c)** 1 : 2 : 1 **(d)** 1 : 2 : 2

10. Assertion (A): Carbon tetrachloride has a lower melting point than sodium chloride. Reason (R): It is a covalent compound with weak forces between molecules. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the mass of water formed when 4 g of hydrogen reacts with 40 g of oxygen. Which reactant is left over and how much? 2

12. Write the electronic configurations of Na^{+} and F^{-} and say what they have in common. 2

13. How many formula units are in 117 g of NaCl relative to 58.5 g? 2

14. An oxide of copper contains 80 per cent copper by mass. Find its empirical formula (Cu = 64, O = 16). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A compound contains 1.6 g of oxygen for 1 g of sulphur (S = 32, O = 16). Find the atoms ratio and the formula. 3

16. Explain why a compound with ions carrying greater charge such as MgO has a higher melting point than NaCl. 3

17. Determine the molecular formula of a compound with empirical formula $\mathrm{CH_2O}$ and molecular mass 60 u. 3

OR

On heating 24.5 g of potassium chlorate, 14.9 g of potassium chloride is left. Find the mass of oxygen liberated and the percentage of oxygen in the original sample. 3

18. The total mass of reactants in a closed container was 50 g. After the reaction the mass of the products found was 49.8 g. Suggest two reasons and comment on the law. 3

OR

Explain how the percentage composition shows that water, $\mathrm{H_2O}$, is different from hydrogen peroxide, $\mathrm{H_2O_2}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student burns 2.4 g of magnesium in a crucible and finds the oxide mass to be 4.0 g. She repeats with 6.0 g and gets 10.0 g. Show that the results support the law of constant proportions, find the ratio Mg : O and verify the formula MgO (Mg = 24, O = 16). 5

OR

Explain why equal masses of different elements do not contain equal numbers of atoms. Compare 12 g of carbon with 12 g of magnesium (C = 12, Mg = 24) in terms of the numbers of atoms. 5

20. Explain how Dalton's atomic theory accounted for the laws of chemical combination and which postulates failed later (subatomic particles, isotopes, isobars) and which ideas remain valid. 5

OR

Analyse the data: elements X and Y form two compounds. Compound I: 1 g X with 1.14 g Y. Compound II: 1 g X with 2.28 g Y. Show that the data illustrate the law of multiple proportions and suggest the formulas if I is XY. 5

END OF QUESTION PAPER — Total: 40 Marks